

Cahier des Charges

Pluridisciplinary Project

Smart Home Assistant and Assistive Mobile Robot

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1 Introduction

This project aims to design and implement a comprehensive multidisciplinary assistive system designed to improve the quality of life for individuals with disabilities or mobility impairments. The system comprises two interconnected subsystems: a **Smart Home Assistant** and an **Assistive Mobile Robot**.

Project Objectives:

- Enhance user autonomy and independence
- Improve safety through hazard detection and prevention
- Provide assistive capabilities for daily living activities
- Ensure system reliability and user-friendly operation

The proposed solution integrates embedded systems, wireless communication technologies, sensor networks, and computer vision techniques to create a cohesive assistive environment.

2 Global System Architecture

The system employs a distributed architecture with two cooperative processing units working in tandem:

2.1 Processing Units

- **Raspberry Pi:** Handles high-level processing tasks including:
 - Voice command recognition and processing
 - Wireless communication management
 - Computer vision and image processing
 - User interface coordination
- **Arduino Microcontroller:** Manages real-time control functions:
 - Sensor data acquisition and processing
 - Actuator control (motors, servos, relays)
 - Safety-critical response mechanisms
 - Low-level hardware interfacing

2.2 Communication Infrastructure

- **Primary Protocol:** Wi-Fi (IEEE 802.11)
- **Advantages:**
 - Remote interaction capability
 - Modular system expansion

- Multi-device connectivity
- Standard protocol support
- **Message Protocol:** HTTP/MQTT (implementation-dependent)

3 System Components

3.1 Subsystem 1: Smart Home Assistant

3.1.1 Overview and Objectives

The Smart Home Assistant provides intelligent control and monitoring of the home environment with a focus on accessibility and safety.

Core Functions:

- Voice-activated control of home devices and appliances
- Security features with password-based access control
- Continuous hazard monitoring (gas leaks, fire detection)
- Multi-modal user alerts (audio and visual feedback)
- Energy-efficient device management

3.1.2 Functional Requirements

Device Control The system shall execute the following commands:

- Lighting control (on/off, dimming where applicable)
- Door lock/unlock operations via servo mechanism
- Appliance control through relay modules
- Scene activation for multiple device coordination

Security Features

- Password verification system for sensitive operations
- Maximum of four consecutive password attempts
- Automatic alarm activation upon repeated failed attempts
- Intrusion alert through buzzer and visual indicators
- System lockdown prevention with emergency reset capability

3.1.3 Safety Monitoring System

Gas Detection

- **Sensor Model:** MQ-2 Gas Sensor
- **Detection Capabilities:**
 - Liquefied Petroleum Gas (LPG)
 - Combustible smoke particles
 - Methane and other combustible gases
- **Threshold Configuration:** Adjustable based on environment
- **Response Time:** < 5 seconds upon detection

Fire Detection

- **Sensor Type:** Infrared Flame Sensor
- **Detection Principle:** Infrared radiation from flames (760-1100nm)
- **Detection Range:** Configurable based on installation location
- **False Positive Mitigation:** Requires sustained signal presence

Alert Mechanisms When hazards are detected, the system activates:

- Continuous audible alarm via buzzer module
- LCD display message indicating:
 - Type of hazard detected
 - Recommended user actions
 - System status
- Optional: Remote notification capability (future enhancement)

System Reset

- Dedicated hardware push button for emergency reset
- Clears alarm state and resets password attempt counter
- Returns system to normal monitoring mode

3.2 Subsystem 2: Assistive Mobile Robot

3.2.1 Overview and Objectives

The Assistive Mobile Robot extends the system's capabilities by providing physical assistance through object retrieval and manipulation.

Primary Functions:

- Indoor navigation and obstacle avoidance
- Object detection and identification
- Autonomous or manual object retrieval
- Pick-and-place operations for lightweight objects
- Real-time video feedback to user interface

3.2.2 Mechanical Design

Mobile Platform

- **Configuration:** Four-wheel drive platform
- **Chassis:** Lightweight, rigid construction
- **Propulsion:** DC motors with independent control
- **Turning Mechanism:** Differential steering
- **Ground Clearance:** Sufficient for indoor surfaces and minor obstacles

Robotic Arm

- **Design:** 4-DOF arm
- **Actuation:** Servo motors for each joint
- **Degrees of Freedom:** 3-4 DOF (base rotation, shoulder, elbow, optional wrist)
- **Gripper:** Servo-actuated parallel jaw gripper
- **Payload Capacity:** Approximately 50-100g for lightweight objects

3.2.3 Control Architecture

The robot employs a hierarchical control structure:

- **High-Level Controller (Raspberry Pi):**
 - Motion planning and navigation
 - Computer vision processing
 - User interface management
 - Wi-Fi communication

- Decision-making algorithms
- **Low-Level Controller (Arduino):**
 - Motor speed and direction control
 - Servo position control
 - Sensor data acquisition
 - Real-time motor feedback
 - Safety limit enforcement

3.2.4 Motor Control System

Motors and Driver

- **Motor Type:** DC Gear Motors (6V or 12V)
- **Motor Driver:** TB6612FNG Dual Motor Driver
- **Driver Features:**
 - Dual H-bridge configuration
 - PWM speed control
 - Direction control
 - Built-in protection circuits
- **Control Signals:** PWM from Arduino for speed regulation

3.2.5 Power Management

- **Primary Power:** Rechargeable battery pack (Li-ion or Li-Po recommended)
- **Voltage Regulation:** Separate regulators for:
 - Motor driver (6V)
 - Arduino (5V)
 - Raspberry Pi (5V, 2.5A minimum)
 - Servos (5V)
- **Battery Monitoring:** Voltage sensing for low-battery warnings
- **Runtime:** Target of 30-60 minutes continuous operation

3.2.6 Perception and Vision System

Camera Configuration

- **Camera Module:** Raspberry Pi Camera Module v2 or compatible
- **Resolution:** 1080p for object detection
- **Mounting:** Pan/tilt servo mount (optional enhancement)
- **Field of View:** Wide-angle for comprehensive scene coverage

Object Detection Strategy Initial Implementation:

- Color-based segmentation for object identification
- Target objects: Three distinctly colored cubes (e.g., red, blue, green)
- OpenCV-based image processing pipeline
- HSV color space for robust color detection

Future Enhancements:

- Machine learning-based object recognition
- Expanded object database
- Natural language object descriptions
- Depth sensing for distance estimation

3.2.7 Operational Modes**Manual Control Mode (Initial Implementation)**

- User controls robot via mobile or web application
- Live video streaming from onboard camera
- Direct control of:
 - Robot movement (forward, backward, turn)
 - Arm joint positions
 - Gripper open/close
- Real-time feedback display

Semi-Autonomous Mode (Future Enhancement)

- User selects target object via interface
- Robot autonomously navigates to object
- Automatic object grasping sequence
- Returns to designated location

Fully Autonomous Mode (Advanced Enhancement)

- Line-following navigation between predefined rooms
- Voice-command object retrieval
- Obstacle detection and avoidance
- Multi-object task sequencing

4 System Integration and Communication

4.1 Communication Protocol

Both subsystems communicate via Wi-Fi using a client-server architecture:

- **Server:** Central Raspberry Pi (Smart Home) or dedicated server
- **Client:** Mobile Robot Raspberry Pi
- **Protocol Options:**
 1. HTTP REST API for command-response operations
 2. MQTT for real-time messaging and status updates

4.2 User Interface

- **Primary Interface:** Dashboard application (Web/Android/iOS)
- **Interface Features:**
 - Voice command input
 - Manual device control
 - System status monitoring
 - Alert notifications

5 Project Constraints

5.1 Development Strategy

Phased Implementation Approach:

1. **Phase 1 - Core Functionality:**
 - Basic voice command recognition
 - Simple device control (lights, door)
 - Gas and fire detection with alarms
 - Manual robot control via interface
2. **Phase 2 - Enhanced Features:**
 - Color-based object detection
 - Robot arm control and object grasping
 - Improved user interface
 - System integration and communication
3. **Phase 3 - Advanced Capabilities (Optional):**

- Semi-autonomous navigation
- Machine learning object recognition
- Multi-room operation
- Remote monitoring and control

6 Testing and Validation

6.1 Subsystem Testing

- **Smart Home Assistant:**
 - Voice command accuracy (>90% target)
 - Security system reliability (password verification)
 - Hazard detection response time (<5 seconds)
 - False alarm rate minimization
- **Assistive Robot:**
 - Navigation accuracy and repeatability
 - Object detection success rate (>85% target)
 - Grasping reliability for target objects
 - Battery runtime verification